DesignJournal Car

DATA VISUALIZATION

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Homework 5



Concept statement

This project was inspired by the desire to combine data visualization with creative and engaging art forms.

Lions were used for multiple purposes in this project. Lions are powerful and charismatic animals that attract people's attention. Each lion represents a car, the shape of which is determined by the car's data attributes such as miles per gallon (MPG), horsepower, number of cylinders, and year of manufacture. This allows viewers to quickly understand the relationships and differences between different cars at a glance.

Adding animations to lions adds extra interactivity and vitality. Blinking and slight movements mimic real behavior, making the visualizations more attractive and relatable. In addition, the interactivity provided by mouse hover and click events enables users to explore the data more deeply and discover the details of each car.



Generative rules

Dataset

The dataset used in this project is a CSV file. It contains information about different cars, with columns such as model, MPG cylinders, horsepower, weight, year, and origin. Each row in the dataset represents a single car, and the values in these columns are used to visually represent the cars in the form of lions.

cars

model	MPG	cylinders	horsepower	weight	year	origin
chevrolet chevelle malibu	18	8	130	3504	70	US
buick skylark 320	15	8	165	3693	70	US
plymouth satellite	18	8	150	3436	70	US
amc rebel sst	16	8	150	3433	70	US
ford torino	17	8	140	3449	70	US
ford galaxie 500	15	8	198	4341	70	US
chevrolet impala	14	8	220	4354	70	US
plymouth fury iii	14	8	215	4312	70	US
pontiac catalina	14	8	225	4425	70	US
amc ambassador dpl	15	8	190	3850	70	US
dodge challenger se	15	8	170	3563	70	US
plymouth 'cuda 340	14	8	160	3609	70	US
chevrolet monte carlo	15	8	150	3761	70	US
buick estate wagon (sw)	14	8	225	3086	70	US
toyota corona mark ii	24	4	95	2372	70	Japan
plymouth duster	22	6	95	2833	70	US
amc hornet	18	6	97	2774	70	US
ford maverick	21	6	85	2587	70	US
datsun pl510	27	4	88	2130	70	Japan
volkswagen 1131 deluxe sedan	26	4	46	1835	70	Europe
peugeot 504	25	4	87	2672	70	Europe
audi 100 ls	24	4	90	2430	70	Europe

Visualization Strategy

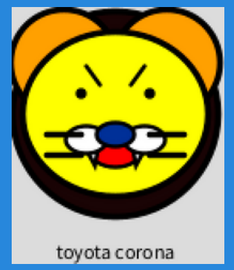
- The visualization presents an overview of the car dataset by representing each car as a lion in a grid layout. The entire grid of lions gives the viewer a quick sense of the number of cars in the dataset.The color of each lion, determined by the car's manufacturing year。.
- When the user hovers the mouse over a lion, detailed information about the corresponding car is displayed near the mouse pointer.
- The relationships between different cars are visually represented through the characteristics of the lions. For example, cars with similar MPG values have similar - shaped eyebrows, indicating a relationship between fuel efficiency and this visual feature. Cars with the same number of cylinders have the same pattern of whiskers, making it easy for the viewer to group and compare cars based on this attribute. The color gradient based on the manufacturing year also shows the temporal relationship between cars.

Generative rules

Specific Mapping Methods

Color

The color of the lion's body is mapped to the car's manufacturing year. A darker colorrepresents earlier years and a red - tinted color represents later years. The map function is used to interpolate between these two colors based on the year value. And the colors of lions' faces represent the country of origin (US,Japan,Europe).



Shape and Size

The size of the lion's ears is determined by the car's weight. A larger weight value results in larger ears.



Facial Features

Eyebrows: The shape and position of the eyebrows are affected by the car's MPG value. Higher MPG values result in different eyebrow shapes.

Whiskers: The number and pattern of whiskers are determined by the number of cylinders in the car. Different cylinder counts (3, 4, 5, 6, 8) have distinct whisker patterns, making it easy to identify and compare cars based on this attribute.

Teeth: Teeth represent the horsepower.

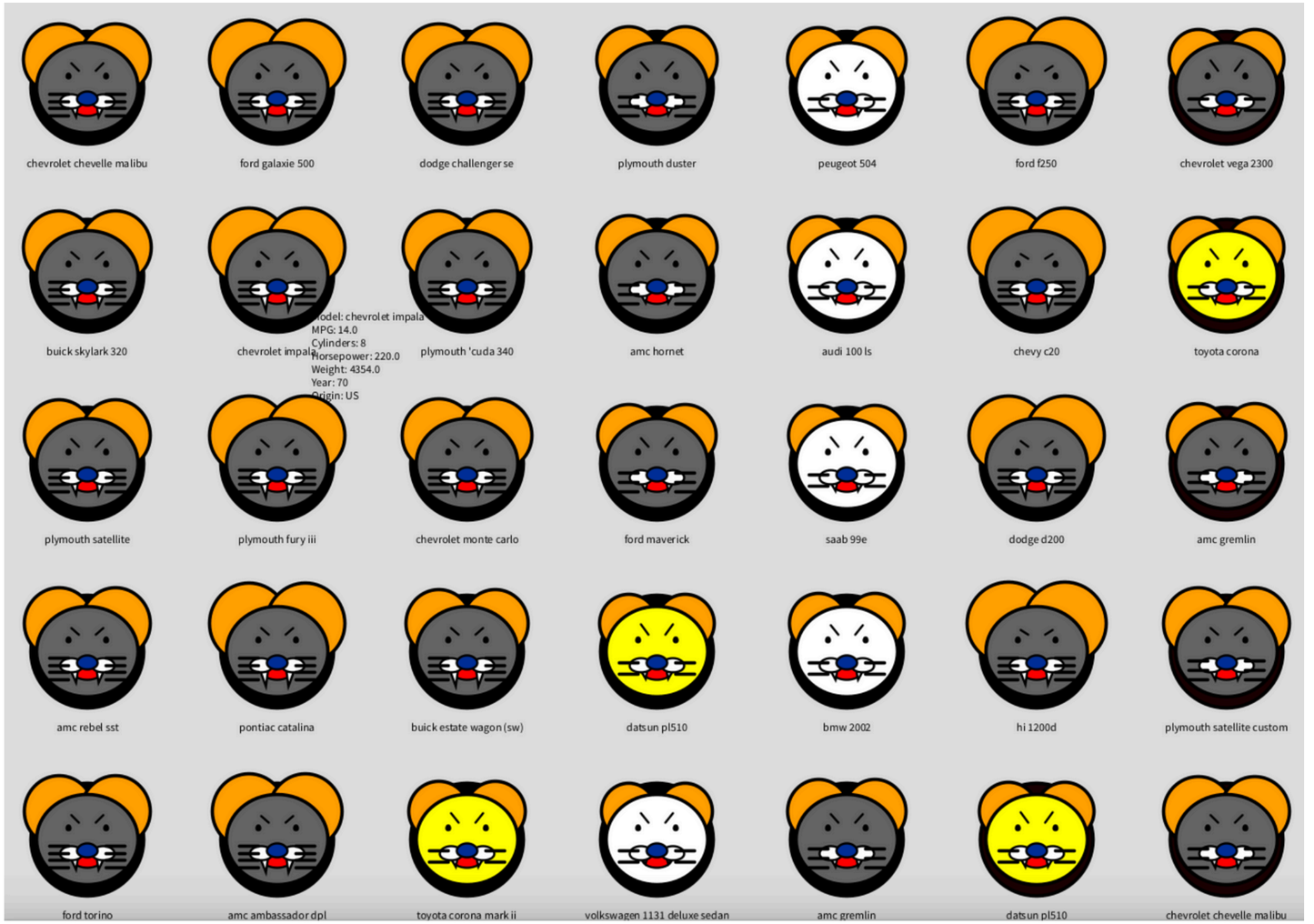


Visualization Strategy

- By using lions to represent cars, the project moves away from the typical dry and technical presentation of automotive data. The animations, such as blinking eyes and gentle movements, add a sense of liveliness and playfulness, making the viewing experience more enjoyable.
- Creative Representation: Using lions as a metaphor for cars is a unique design choice that immediately captures the viewer's attention.
- Interactivity: When the mouse hovers over a lion, detailed information is displayed. If you click the mouse,and that will turn over to the next page. These interactions allow the user to actively engage with the data and discover more about each car.
- Animation: The blinking eyes and gentle movements of the lions enhance the overall experience, making the visualization more dynamic and memorable.



Outcome


chevrolet chevelle malibu
ford galaxie 500
dodge challenger se
plymouth duster
peugeot 504
ford f250
chevrolet vega 2300
buick skylark 320
chevrolet impala
plymouth 'cuda 340
amc hornet
audi 100 ls
chevy c20
toyota corona
plymouth satellite
plymouth fury iii
chevrolet monte carlo
ford maverick
saab 99e
dodge d200
amc gremlin
amc rebel sst
pontiac catalina
buick estate wagon (sw)
datsun pl510
bmw 2002
hi 1200d
plymouth satellite custom
ford torino
amc ambassador dpl
toyota corona mark ii
volkswagen 1131 deluxe sedan
amc gremlin
datsun pl510
chevrolet chevelle malibu
Model: chevrolet impala
MPG: 14.0
Cylinders: 8
Horsepower: 220.0
Weight: 4354.0
Year: 70
Origin: US